Griffin Kriebel

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Education:

University of California, Davis — *Bachelor of Science, Biological Systems Engineering* **Expected Dec 2026**
ABET accredited coursework in Mechanical, Electrical, and Biotechnical Engineering

Experience:

Student Researcher, *Agnetix LED Liquid Cooling Heat Recycler* — Davis, CA **Sep 2025 - June 2026**

- UC Davis Senior Design Project partnered with Agnetix using liquid cooling and heat transfer of LEDs to increase energy efficiency and crop productivity for indoor vertical farms.
- Designed and implemented PLC automated temperature control where root solution is kept between 25-30℃ using recycled LED heat.
- Removed excess heat from the system using a 1800W heat exchanger designed for peak thermal demand.
- Increased energy efficiency from 72% to 94% with an average heat recovery of 948W equivalent to 6.81℃.

Agricultural Robotics Intern, *Plant Sciences Inc.* — Manteca, CA **June 2024 - July 2024**

- Collected field data to evaluate robotic and computer vision performance in strawberry flower elimination.
- Proposed design modifications to improve mechanical reliability and operational efficiency.
- Digitized commercial farm maps using Microsoft Excel to streamline field logistics.

Orchard Maintenance Intern, *Weaver's Orchard* — Morgantown, PA **June 2023 - Sep 2023**

- Designed and implemented drip irrigation systems for fruit and vegetable fields.
- Operated and maintained agricultural machinery across a 128-acre orchard.
- Performed crop care tasks including disease and pest management, thinning, and harvesting.

Projects:

UC Davis Renewable Grid Design, UC Davis Sustainable Power Generation **March 2025 - June 2025**

- Collaborated to develop a cost-optimized hybrid renewable energy system using NREL's System Advisor Model (SAM) to reduce grid dependency and meet annual energy demand.
- Processed and analyzed hourly annual energy data in Excel to identify the optimal wind-to-solar ratio, reducing supplemental grid electricity use by 70% and LCOE to 7.24¢/kWh over 25 years.

VHS to Digital Camcorder Conversion, Personal Project | **GrabCAD Model** **Aug 2025 - Sep 2025**

- Commissioned by a client to create a VHS camera to record in digital format for concert videography.
- Designed a custom 12V 2600mAh battery pack and integrated a miniature DVR for analog-to-digital video conversion with a battery life of 6 hours.
- Modified the 10-pin camera connector to accommodate DVR output wires and power distribution.

Bike to E-bike Conversion, Personal Project | **GrabCAD Model** **Sep 2024 - Jan 2025**

- Designed and 3D-printed adaptive motor mounts for electric bike conversion with top speeds of 20 mph.
- Applied iterative design process and tested mechanical-electrical integration to achieve a 15 mile range.

Vertical Axis Wind Turbine, Personal Project | **GradCAD Model** **Dec 2024 - Jan 2025**

- Designed and assembled a 3D-printed wind turbine prototype using Fusion 360.
- Analyzed performance metrics including wing swept area, rated power, and power curve to evaluate efficiency.

Solar Dehydrator, UC Davis EBS 001 | **GrabCAD Model** **Sep 2024 - Dec 2024**

- Collaborated in a team of four to design, build, and test a food dehydration system.
- Analyzed data on moisture loss and presented efficiency findings in a final report.

Technical Skills: Fusion360, MATLAB, Solidworks, Python, COMSOL, 3D Printing, Metal Fabrication
Professional Skills: Communication, Leadership, Technical Writing, Organized, Creative, Adaptable

Griffin Kriebel - Engineering Portfolio



https://griffinkriebel.com/